

JAYDEEP GOSWAMI

Backend Software Engineer

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PROFESSIONAL SUMMARY

Backend Software Engineer with **3.10+ years** building distributed, cloud-scale systems — including a network platform managing **20,000+ devices** with **40% API latency reduction** and **45% server load reduction**. Currently building Jobspiq, an AI-native job platform, as sole architect and engineer — covering semantic search, multi-portal crawling, and production deployment. Seeking senior backend or founding engineer roles where I can own systems end-to-end and grow toward technical leadership.

SKILLS

- **Languages:** Python, JavaScript
- **Frameworks:** Django REST Framework, FastAPI, Flask
- **Databases:** PostgreSQL, MySQL, Cassandra, Redis, MongoDB
- **Messaging:** RabbitMQ, Celery, Redis Pub/Sub
- **DevOps & Cloud:** AWS, GCP, Docker, Kubernetes, Jenkins, CI/CD pipelines, OCI
- **AI & LLM Integration:** LLM API integration, Prompt engineering, Context injection, Token management, System prompt design, pgvector, Semantic search, Cosine similarity
- **Data Processing:** Pandas, NumPy, Pydantic
- **Auth & Security:** JWT, OAuth2
- **Testing & Tools:** Pytest, Postman, Git, GitLab, Jira, Linux
- **Practices:** Microservices, REST API Design, ORM (SQLAlchemy), Agile
- **Real-time systems:** Multiprocessing, Ring buffer design, CPU affinity, Memory bandwidth profiling

EXPERIENCE

VVDN Technologies — Software Engineer

Gandhinagar, Gujarat | October 2022 – December 2025

HFCL – Element Management System

October 2022 – Present

Python · Pandas · RabbitMQ · Redis · Celery · Django · REST APIs · Cassandra · Kubernetes · Docker

- Architected backend services for a cloud-scale network EMS platform managing 20000+ network devices, using Python, Django REST Framework, and Cassandra for high-throughput device telemetry ingestion.
- Built async event pipelines using RabbitMQ and Celery for decoupled, fault-tolerant processing of device configuration and alert events at scale.
- Improved system performance by 20% through Pandas-based batch data processing and Python threading for parallel execution across device polling cycles.
- Designed a Redis caching layer and optimized Cassandra bulk-write pipelines, cutting redundant DB calls by 30% and reducing server load by 45% under peak traffic.
- Deployed and maintained services on Kubernetes with Docker containerization, supporting zero-downtime rollouts in production.

MEsS – Manufacturing Execution System

October 2023 – April 2024

Python · FastAPI · Redis · Celery · PostgreSQL · Docker · Linux · Postman

- Built a real-time PCB traceability system using FastAPI and PostgreSQL, tracking board lifecycle across 8+ SMT assembly stations with per-stage pass/fail logging.

- Designed and managed PostgreSQL schemas for structured manufacturing and traceability data, enabling compliance-ready audit trails.
- Led requirement-gathering sessions with production and quality teams, translating shop-floor workflows into actionable sprint features in Agile cycles.
- Containerized all services with Docker to ensure environment consistency across development, staging, and production.

S3DT — Real-Time Camera Acquisition System

January 2026 – Present

Multiprocessing · FFmpeg · NumPy · FLIR SDK · Linux · Tkinter · FastAPI

- Architected a high-performance 8-camera acquisition system capable of recording synchronized 291–300 FPS video streams with zero frame loss through multiprocessing, shared-memory pipelines, and optimized capture strategies.
- Redesigned the complete desktop acquisition software by migrating from a Tkinter-based application to a FastAPI web architecture, enabling browser-based control, monitoring, and simplified deployment.
- Built a real-time streaming pipeline using WebSockets, allowing low-latency live previews of all connected cameras over the local network while recording high-speed video simultaneously.

EARLY EXPERIENCE

Trainee – Software Engineer | VVDN Technologies, Ahmedabad | June 2022 – September 2022

AWS Intern | Zummit Infolabs (Remote) | January 2022 – May 2022

PROJECTS

Jobspiq — (jobspiq.in)

January 2026 – Present

AI-native job aggregation platform with semantic matching, multi-portal crawling, and full-stack solo deployment.

FastAPI · PostgreSQL · pgvector · Elasticsearch · Celery · Redis · Docker · OCI · Playwright

- Semantic matching engine using BAAI/bge-base-en-v1.5 (768D) embeddings with pgvector cosine similarity + Elasticsearch pre-filtering against user Professional DNA profiles.
- Multi-portal crawlers (Naukri, LinkedIn, Indeed, Wellfound, Cutshort) with SHA-256 deduplication, Playwright enrichment, and Celery Beat scheduling across 4 task queues.
- Full auth system — password login, OTP signup, and password reset backed by Redis TTL. Deployed on OCI ARM64 with Docker Compose and Nginx reverse proxy.

DisciplineOS — Open source: github.com/JaydeepGIO/DisciplineOS

January 2026 – March 2026

Open-source productivity OS with a discipline scoring engine and LLM-powered coaching layer.

Python · FastAPI · Redis · Celery · PostgreSQL · Docker · LLM APIs · Postman

- Discipline scoring engine using async Celery workers — automated streak tracking, habit scoring, and background report generation.
- LLM coaching layer (DisciplineCoach) with context injection, token budgeting, sliding window memory, and grounding patterns to prevent hallucination.

EDUCATION

Bachelor of Engineering | Gujarat Technological University, Gandhinagar | June 2022 | CGPA: 8.22/10